HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed **Python** scripts to automate data extraction and transformation for analytical reporting, processing approximately **1.5 TB** of raw data weekly.
- Designed and implemented **Java Spring Boot** REST APIs to facilitate data exchange between internal services, handling over **5,000 requests per minute** during peak load.
- Managed production incidents for enterprise applications using **Splunk** and **Dynatrace**, consistently meeting **P1/P2 SLA targets** through rapid root cause analysis and resolution.
- Constructed real-time data pipelines using **Python** and **Pandas** on **AWS Lambda** to ingest and process streaming financial transaction data, reducing data latency by **40%**.
- Trained a **scikit-learn** classification model on a **~250,000-row** transaction dataset to identify anomalous patterns, achieving an **AUC of 0.89** on held-out validation data.
- Containerized applications using **Docker** and deployed them to **AWS EC2** instances, integrating with **Jenkins** for automated build and deployment pipelines.
- Collaborated with trading strategy development teams to gather technical requirements and translate them into actionable development tasks for quantitative strategy analysis projects.

SKILLS

Languages – Python, Java, SQL, R

Frameworks/Libraries – Spring Boot, Pandas, NumPy, scikit-learn, FastAPI

Cloud/DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools/Platforms – Splunk, Dynatrace, Jira, Confluence, Power BI, Tableau

Concepts – REST APIs, Data Pipelines, ETL, CI/CD, Agile, Quantitative Analysis, Machine Learning, Statistical Analysis

PROJECTS

Quantitative Strategy Analysis Pipeline (HCL Technologies)

- Engineered a **Python**-based data pipeline to process and clean large volumes of market data, supporting quantitative strategy analysis workflows.
- Developed **SQL** queries against **PostgreSQL** to extract and aggregate historical trading data, forming the basis for predictive feature generation.
- Implemented automated reporting modules using **Pandas** that summarized key performance indicators and statistical deviations for strategy backtesting, reducing manual reporting effort by **~60%**.

Anomaly Detection Model for Transaction Streams (MSc Project)

- Developed a machine learning model using **scikit-learn** (Random Forest classifier) to detect anomalous financial transactions from a dataset of **~300,000 records**.
- Extracted and engineered relevant features from raw transaction logs, including velocity checks and historical patterns, and evaluated model performance using **AUC (0.91)** and **F1-score (0.48)**.
- Packaged the model inference logic into a **FastAPI** service, enabling real-time scoring with a p95 latency of **~220ms**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)